ORLANDO HEALTH
1414 Kuhl Ave.
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DEPARTMENT GUIDELINE

Guideline Title: **Thoracic Trauma**	Guideline Number: **GUID-3203-T015**

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Revision dates: 07/01/2026	Developed by: Air Care Team
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Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To recognize the patient who has suffered thoracic trauma and treat life-threatening injury immediately. To provide the patient with adequate airway, perfusion, and pain management and transport to a facility with the ability to care for the patient.

II. **PATHOPHYSIOLOGY:** Thoracic trauma is classified as blunt or penetrating. Blunt trauma to the thorax is associated with vehicular collisions, falls, and assaults, whereas penetrating trauma occurs as a result of gunshot wounds, stab wounds, and impalement. Life threatening thoracic injuries may include: airway obstruction, tension pneumothorax, massive hemo-thorax, open pneumothorax, flail chest, cardiac tamponade, aortic rupture/transection, myocardial rupture, myocardial contusion, pulmonary contusion, tracheobronchial disruption, diaphragmatic disruption. A decrease in the blood volume, failure to ventilate the lungs, ventilation perfusion mismatches or pressure changes within the intrapleural space are contributing causes of hypoxia.

III. **DEFINITIONS:**

IV. **DEPARTMENT GUIDELINE:**

   a. Conduct primary assessment; major problems identified should be immediately treated.
   b. Assess for and control any external hemorrhage. Stabilize any penetrating objects.
   c. Assess and manage airway, breathing, and circulation.
      i. Evaluate for partial/full obstruction, stridor, increased secretions, blood in the airway.
      ii. Assess respiratory rate, effort, breath sounds, position of trachea, external signs of chest trauma, bony or air crepitus upon palpation, paradoxical movement, presence of flail segment and/or presence of JVD.
      iii. Assess SpO2 reading as well as ETCO2 if applicable.
   d. Assess for mechanism of injury, vehicle damage such damage to the steering wheel or steering column, use of restraints, intrusion into the driver or passenger compartment, ejection of patients, height of fall and/or speed at which the pedestrian was struck.
   e. Assess for signs of shock, altered mentation, hypotension, or increased HR, and/or signs of peripheral vascular constriction. Monitor continuous vital signs.

f. If hemodynamic instability, consider blood product transfusion per *GUID3203-PR025 - Emergency Blood Product Administration* or *GUID3203-PR002 Blood Product Administration* with *GUID3203-PR026 – Warming of Blood Products & Intravenous Fluids*.
g. If signs of a tension pneumothorax (increased heart or respiratory rate, decreased SpO2, decreased lung sounds to the affected side, JVD, subcutaneous emphysema, hypotension, or tracheal deviation), perform a needle thoracostomy per *GUID3203-PR017- Needle Thoracostomy*.
h. If clinical signs of an open pneumothorax, place a sterile occlusive dressing, large enough to overlap the wound edges, that is taped securely on 3 sides. If signs of a tension pneumothorax develop, release occlusive dressing before performing a needle thoracostomy.
i. Stabilize any noted flail segments. Consider positive pressure ventilation
j. Provide adequate analgesics per *GUID3203-G005 – Analgesia and Sedation Management*.
k. If interfacility transport
    i. Assess initial chest radiograph for signs of hemothorax/pneumothorax.
    ii. If chest tube is in place, maintain closed chest tube drainage system. When feasible, maintain suction per *GUID3203-PR005 – Chest Tube Management*.
    iii. If *confirmed* blunt aortic injury, consider maintaining HR < 100 bpm and SBP 100-120 mmHg with **Esmolol bolus 500 mcg/kg/min IV** over 1 minute followed by **Esmolol infusion at 50 mcg/kg/min**. Titrate every 5 minutes by repeating 500 mcg/kg/min bolus over 1 minute and then increasing infusion by 50 mcg/kg/min. Maximum dose 300 mcg/kg/min
        1. If esmolol unavailable, **Labetalol 10-20 mg IV** over 2 minutes. May repeat every 10 minutes until goal blood pressure obtained or maximum total dose of 300mg. Avoid if HR < 60 bpm.
    iv. Obtain all pertinent paperwork, X-rays, and CT scans.
    v. Consider obtaining a 12-lead ECG.
l. In the setting of traumatic cardiac arrest with suspected chest trauma, consider *bilateral pleural decompressions* as part of the resuscitation efforts.

V. **DOCUMENTATION:**
    a. EMS Charts

VI. **REFERENCES:**
    a. Fabian, T.C., Davis, K.A., Gavant, M.L., et al. (1998). Prospective study of blunt aortic injury: helical CT is diagnostic and antihypertensive therapy reduces rupture. *Ann Surg*, 277, 666.